

Problem 278: Bricks-timator

Difficulty: Easy

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Originally Published: Code Quest 2026

Problem Background

Construction crews often need to order the correct number of bricks before a project begins. Ordering too few bricks causes delays, while ordering too many wastes money and storage space. Your task is to write a program that, given the size of the bricks, the uniform wall height, the number of walls, and the length of each wall, computes the minimum number of bricks that must be purchased to complete the task.

Problem Description

Please note this problem overrules the general guidelines for rounding numbers provided in the Reference Materials.

You'll be provided with the length and height of each brick, as well as the dimensions of each wall that needs to be built with those bricks. All measurements will be provided in the same units.

Each wall is one brick deep (so thickness is ignored for this problem); additionally, each brick will be placed so its length is parallel to the wall's length and its height is parallel to the wall's height. In other words, you won't need to rotate the bricks in order to build each wall. Bricks *can* be cut in order to fit the dimensions of the wall, however, so the number of bricks required for a wall is just the wall's area divided by a brick's area.

The construction project is for a one-story building, so all walls will be the same height, but will have differing lengths. You'll need to determine the total number of bricks needed to complete all of the walls, assuming no overlap between them. **Round your final result upward** to ensure you order enough bricks to complete the project.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include a single line containing the following values, separated by spaces:

- The dimensions of a brick in the form "LxH", where L is a positive integer representing the length of a single brick, x is a literal 'x' character, and H is a positive integer representing the height of a single brick.
- A positive integer representing the height of all walls to be constructed.

- A positive integer, **N**, representing the number of walls to be constructed.
- **N** positive integers, each representing the length of a single wall to be constructed.

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6x2 46 5 22 22 22 22 12

4x4 32 4 12 16 20 24

Sample Output

For each test case, your program must print a line containing an integer representing the minimum number of bricks required to complete the construction project. **Round all non-integer results upward to the next highest integer.**

384

144